

- **Employee Dataset**

DROP Database if Exists company _ db;

Create Database Company _ db;

Use Company _ db;

Create table Employees (
Emp _id INT primary Key,
name varchar (50),
department _id varchar (50),
salary int
);

Insert INTO Employees values

(101,'Abhishek','D01',62000),

(102,'Shubham','D01',58000),

(103,'Priya','D02',67000),

(104,'Rohit','D02',64000),

(105,'Neha','D03',72000),

(106,'Aman','D03',55000),

(107,'Ravi','D04',60000),

(108,'Sneha','D04',75000),

(109,'Kiran','D05',70000),

(110,'Tanuja','D05',65000);

Select * From Employees;

<i>Emp _id</i>	<i>name</i>	<i>Department _id</i>	<i>salary</i>
101	Abhishek	D01	62000
102	Shubham	D01	58000
103	Priya	D02	67000
104	Rohit	D02	64000
105	Neha	D03	72000
106	Aman	D03	55000
107	Ravi	D04	60000
108	Sneha	D04	75000
109	Kiran	D05	70000
110	Tanuja	D05	65000

- **Department Dataset**

Use Company _db;

Create table Department (

Department _id Varchar (50) primary Key,

Department _name Varchar (50),

location Varchar (50)

);

Insert INTO Department values

('D01','Sales','Mumbai'),

('D02','Marketing','Delhi'),

('D03','Finance','Pune'),

('D04','HR','Bengaluru'),

('D05','IT','Hyderabad');

select * from Department;

<i>Department_id</i>	<i>Department_name</i>	<i>location</i>
D01	Sales	Mumbai
D02	Marketing	Delhi
D03	Finance	Pune
D04	HR	Bengaluru
D05	IT	Hyderabad

- **Sales Dataset**

Use Company _db;

DROP TABLE IF EXISTS Sales;

Create table Sales (

Sale _id INT primary Key,

Emp _id INT,

Sale _amount INT,

Sale _date Date

);

Insert INTO Sales values

(201,101,4500,'2025-01-05'),

(202,102,7800,'2025-01-10'),

(203,103,6700,'2025-01-14'),

(204,104,1200,'2025-01-20'),

(205,105,9800,'2025-02-02'),

(206,106,10500,'2025-02-05'),

(207,107,3200,'2025-02-09'),

(208,108,5100,'2025-02-15'),

(209,109,3900,'2025-02-20'),

(210,110,7200,'2025-03-01');

select * from Sales;

<i>Sale_id</i>	<i>Emp_id</i>	<i>Sale_amount</i>	<i>Sale_date</i>
201	101	4500	2025-01-05
202	102	7800	2025-01-10
203	103	6700	2025-01-14
204	104	1200	2025-01-20
205	105	9800	2025-02-02
206	106	10500	2025-02-05
207	107	3200	2025-02-09
208	108	5100	2025-02-15
209	109	3900	2025-02-20
210	110	7200	2025-03-01

- **Basic Level**

1. Retrieve the names of employees who earn more than the average salary of all employees.

```
SELECT name  
FROM Employees  
WHERE salary > (SELECT AVG (salary) FROM Employees);
```

<i>Name</i>
Priya
Neha
Sneha
Kiran
Tanuja

2. Find the employees who belong to the department with the highest average salary.

```
SELECT name
FROM Employees
WHERE department _id = (
    SELECT department _id
    FROM Employees
    GROUP BY department _id
    ORDER BY AVG (salary) DESC
    LIMIT 1
);
```

<i>Name</i>
Ravi
Sneha

3. List all employees who have made at least one sale.

Select Name

From Employees

Where Emp _id IN (Select Distinct Emp _id From Sales);

<i>Name</i>
Abhishek
Shubham
Priya
Rohit
Neha
Aman
Ravi
Sneha
Kiran
Tanuja

4. Find the employee with the highest sale amount.

Select name

From Employees

Where Emp _id = (

Select Emp _id

From Sales

Order by Sale _amount DESC

Limit 1

);

<i>Name</i>
Aman

5. Retrieve the names of employees whose salaries are higher than Shubham's salary.

Select Name

From Employees

Where Salary > (

Select Salary From Employees Where name = 'Shubham'

);

<i>Name</i>
Abhishek
Priya
Rohit
Neha
Ravi
Sneha
Kiran
Tanuja

- **Intermediate Level**

1. Find employees who work in the same department as Abhishek.

```
SELECT name
FROM Employees
WHERE department_id = (
    SELECT department_id
    FROM Employees
    WHERE name = 'Abhishek'
);
```

<i>Name</i>
Abhishek
Shubham

2. List departments that have at least one employee earning more than ₹60,000.

```
SELECT department _name
FROM Department
WHERE department _id IN (
    SELECT DISTINCT department _id
    FROM Employees
    WHERE salary > 60000
);
```

<i>Department _name</i>
Sales
Marketing
Finance
HR
IT

3. Find the department name of the employee who made the highest sale.

```
SELECT department _name
FROM Department
WHERE department _id = (
    SELECT department _id
    FROM Employees
    WHERE emp _id = (
        SELECT emp _id
        FROM Sales
        ORDER BY sale _amount DESC
        LIMIT 1
    )
);
```

<i>Department _name</i>
Finance

4. Retrieve employees who have made sales greater than the average sale amount.

```
SELECT DISTINCT e.name
```

```
FROM Employees e
```

```
JOIN Sales s ON e. emp _id = s. emp _id
```

```
WHERE s. sale _amount > (SELECT AVG (sale _amount) FROM Sales);
```

<i>Name</i>
Shubham
Priya
Neha
Aman
Tanuja

5. Find the total sales made by employees who earn more than the average salary.

```
SELECT SUM (s. sale _amount) AS total _sales  
FROM Sales s  
WHERE s. emp _id IN (  
    SELECT emp _id  
    FROM Employees  
    WHERE salary > (SELECT AVG (salary) FROM Employees)  
);
```

<i>Total _sales</i>
32700

- **Advanced Level**

1. Find employees who have not made any sales.

```
SELECT name
```

```
FROM Employees
```

```
WHERE emp_id NOT IN (SELECT emp_id FROM Sales);
```

<i>Name</i>

2. List departments where the average salary is above ₹55,000.

```
SELECT department _name
FROM Department
WHERE department _id IN (
    SELECT department _id
    FROM Employees
    GROUP BY department _id
    HAVING AVG (salary) > 55000
);
```

<i>Department _name</i>
Sales
Marketing
Finance
HR
IT

3. Retrieve department names where the total sales exceed ₹10,000.

```
SELECT department _name
FROM Department
WHERE department _id IN (
    SELECT e. department _id
    FROM Employees e
    JOIN Sales s ON e. emp _id = s. emp _id
    GROUP BY e. department _id
    HAVING SUM (s. sale _amount) > 10000
);
```

<i>Department _name</i>
Sales
Finance
IT

4. Find the employee who has made the second-highest sale.

```
SELECT name
FROM Employees
WHERE emp_id = (
    SELECT emp_id
    FROM Sales
    ORDER BY sale_amount DESC
    LIMIT 1 OFFSET 1
);
```

<i>Name</i>
Neha

5. Retrieve the names of employees whose salary is greater than the highest sale amount recorded.

SELECT name

FROM Employees

WHERE salary > (SELECT MAX (sale _amount) FROM Sales);

<i>Name</i>
Abhishek
Shubham
Priya
Rohit
Neha
Aman
Ravi
Sneha
Kiran
Tanuja